

Group 6: Harshini Padala (hpadala), Shikha Nukala (desigirl), Anvita Gollu (anvitag), Nishita Shah (snishita), Kaelyn Lin (kaelynl)

Project Context, Expected Deliverables, and Scope

1. Project context and user/stakeholder info
 - a. Our aim is to reduce food waste in restaurants by creating a system that efficiently matches restaurants with surplus food with shelters and soup kitchens that need it. This initiative will not only minimize waste but also improve food security by ensuring that safe and consumable food reaches those in need. Our first priority (5/5) is to streamline the donation process by automating matches based on food type, dietary restrictions, and real-time availability in an efficient, single-step method. Additionally, many restaurants already adhere to FDA regulations. By encouraging more restaurants to donate, we enhance the safety and quality of donated food, another top priority for us. As secondary priorities (4/5), restaurants can also receive cost reduction and increased reputation benefits from this application. Restaurants can reduce disposal costs and potentially receive tax incentives for donating food. Restaurants can also improve their public image by demonstrating social responsibility and sustainability efforts. We plan on focusing on two key categories of priorities to maximize the impact we make on them instead of focusing on too many priorities. By integrating regulatory guidance and real-time food matching, our platform will remove logistical barriers and encourage more businesses to participate in food donation, creating a win-win solution for both restaurants and communities in need.
 - b. Our primary users include restaurant owners and managers, who will use the platform to list surplus food, match with shelters, and ensure compliance with FDA regulations. Our users also include homeless shelters and soup kitchens, which will receive and distribute donated food based on dietary needs and availability. Key stakeholders include the FDA and food safety regulatory agencies, which ensure compliance with donation laws. Also local governments and municipalities, which may provide policy support, tax incentives, or logistics assistance for food redistribution. Lastly, competitors, such as other food donation platforms, are who we will compare our application to. This will help us identify areas where we can differentiate ourselves based on the user's needs.
2. Expected deliverables and scope
 - a. Our initial and most important deliverables include a user authentication system to ensure secure access, a basic matching system that connects donors with recipients through filtering preferences, and a contact form to facilitate communication between users and the platform administrators. Additionally, we will provide educational resources covering FDA regulations and donation procedures to guide both donors and recipients in compliance and best practices.
 - b. As potential stretch goals, we aim to implement background check capabilities for added security, mobile design to enhance accessibility, updates on the status of

food donations to keep users informed, and profile information featuring ratings and feedback to build trust and accountability within the platform.

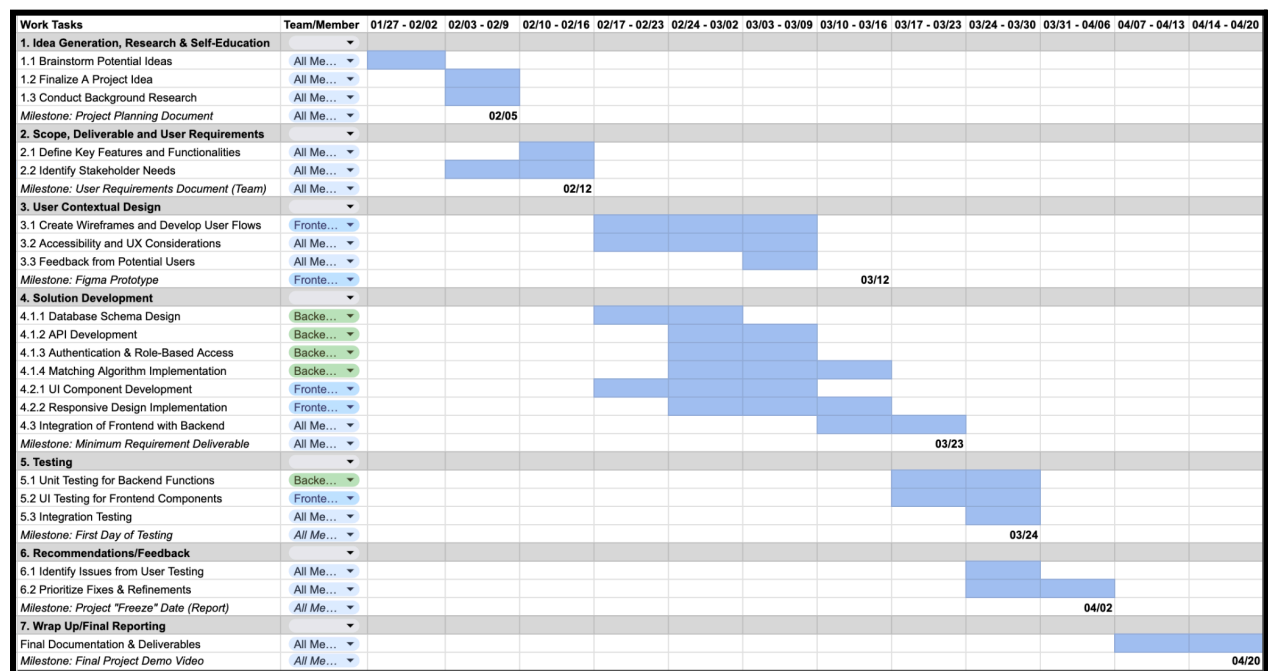
- c. In scope for this project is the development of a website or web-based platform with a focus on user interface (UI) design, the integration of educational content, and the creation of a basic matching system that connects food donors to shelters based on preferences and needs. While these features are our primary focus, secondary priorities include ensuring mobile flexibility, implementing UI updates for every stage of the donation process, and adding notifications for new potential matches. Out of scope for this project are advanced matching algorithms that use weighted preference systems, user account settings, multilingual support, and the inclusion of national clients. These features are either beyond the current project's capacity or not directly relevant to our immediate goals.
- d. The need for this platform is driven by concerning food waste statistics, and through relevant academic literature we aim to highlight how frequently food donations occur and how much of that food actually reaches shelters. To better understand user needs, we've also gathered survey data and feedback from local restaurants and homeless shelters from previous projects, providing insight into their donation experiences and pain points. We are also reviewing existing food donation platforms to identify gaps and opportunities for improvement in our design. Our approach will rely on basic web technologies and user-centered design principles, with matching algorithms based on simple filtering criteria. Additionally, we plan to engage with content experts, including faculty members and professionals experienced in food donation logistics, to ensure the platform aligns with industry best practices and regulatory guidelines.

High-level, Prioritized User Requirements Table with Target Values

User Requirement	Priority (1-5)	Specifications	Evaluation Method	Not-you Requirement
User Authentication	1	Users can securely log in/out; system prevents unauthorized access	Test with backend DB; penetration testing	Yes
Matching Algorithm	1	Every shelter gets at least one restaurant match	Functional testing with sample data	Yes
User Profiles	1	Users can set/edit preferences for donations and requests	Backend testing; UI Validation	Yes
User-friendly interface	1	Simple, easy to navigate, and intuitive user interface	User testing and feedback analysis (not-you)	Yes
Education resources	2	Informational section with donation guidelines and statistics	Content review and user feedback (not-you)	Yes
Email notifications for status tracking	5	Automated email notifications for new matches	Functional testing; email logs	Yes
Real-Time Updates	5	Live dashboard for donation and request status	Performance testing; UI validation	Yes
Feedback form	3	Users can provide feedback on matches	Survey analysis; feedback logs	Yes

1 = highest, 5 = lowest

Our initial project management strategy remains unchanged. Our team structure continues with a designated leader and an equal division of members focusing on backend and frontend development. We have committed to weekly meetings every Monday at 11:30 AM, ensuring we stay aligned and fully prepared for our bi-weekly check-ins with the 497 staff. Below is our updated Gantt chart.



Primary Persona

Frustrations

- Unclear food safety rules & liability concerns
- No easy way to find shelters in real-time
- Time-consuming coordination & logistics

ALEX

"I want to give back to the community, but donating surplus food should be as easy as placing an order online."

RESTUARANT OWNER PERSONA

Needs

- ✓ A simple, fast donation matching system
- ✓ Clear FDA guidelines for safe food donations
- ✓ Reliable tracking and feedback from shelters

Goals

Reduce food waste efficiently

Support the community by donating surplus food

Ensure compliance with food safety regulations

Streamline the donation process with minimal effort